25901262 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCFSQ201 - INTRODUCTION TO FOOD MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Identify the part of eukaryotic cell called the “powerhouse” of the cell. [U] / [CO1]

   a). Ribosome b). Lysosome

   c). Mitochondria d). Golgi apparatus

2. Select the correct covering found in protozoa like Paramecium: [K] / [CO1]

   a). Cell wall b). Capsule

   c). Pellicle d). Cuticle

3. Define pure culture. [U] / [CO2]

   a). Culture containing many species b). Culture contaminated with fungi

   c). Population of cells arising from a single cell d). Mixed microbial population

4. Define physical sterilization. [U] / [CO2]

a). Disinfection only b). Cleaning process

c). Sterilization using physical agents d). Sterilization using chemicals

5. Vibrio cholerae infection is mainly transmitted through [K] / [CO3]

a). Air b). Contaminated water and food

c). Sunlight d). Dust

6. Soft swell has [U] / [CO3]

a). Both ends of the can bulged b). Distorted ends of the can

c). Flat ends of the can d). Dented seams of the can

7. Identify the food which is most susceptible to aflatoxin contamination [K] / [CO3]

a). Fresh vegetables b). Nuts and oilseeds

c). Fish d). Meat

8. Soft rot of vegetables is mainly caused by [U] / [CO3]

a). fungal cellulases b). bacterial pectinolytic enzymes

c). lipid oxidation d). starch breakdown

9. Deterioration of fatty fish produces appreciable amounts of ‘stale fishy’ smell, which is due to _____. [U] / [CO3]

a). Amino acids b). Ammonia

c). Chloramines d). Trimethylamine oxide

10. Putrefaction is mainly associated with degradation of [K] / [CO3]

a). carbohydrates b). fats

c). proteins d). minerals

[1x10 = 10]

Part B

Short Answer

Answer any **4** questions

Each question carries **5** marks

11. State the contributions of Robert Koch and outline Koch's postulates. [U] / [CO1]

12. Describe the different phases of bacterial growth curve [U] / [CO1]

13. Identify the structural components of a typical virus and state the function of each. [U] / [CO1]

14. Describe the principle and procedure of spread plate method. [U] / [CO2]

15. Define putrefaction and explain its significance in meat spoilage. [K] / [CO3]

16. Describe the role of moisture in spoilage of stored cereal grains. [U] / [CO3]

[5x4 = 20]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

17. Recall the methods of sexual reproduction in fungi. [U] / [CO1]

18. Explain any five chemical sterilizating agents and summarize the factors that influence their effectiveness. [U] / [CO2]

19. Explain Vibrio cholerae and Cholera in detail. [U] / [CO3]

20. Describe the spoilage of milk with reference to acid, proteolytic, and gas-forming spoilage. [U] / [CO3]

[10x2 = 20]